

Library Management System - Complete Modules

src/App.js

```
import React, { useState } from "react";
import "./style.css";

function App() {
  const [books, setBooks] = useState([
    {
      id: 1,
      title: "Python Basics",
      author: "John",
      quantity: 5,
      borrowed: false,
      reserved: false
    },
    {
      id: 2,
      title: "React Guide",
      author: "David",
      quantity: 3,
      borrowed: false,
      reserved: false
    }
  ]);

  const [search, setSearch] = useState("");
  const [newBook, setNewBook] = useState({
    title: "",
    author: "",
    quantity: ""
  });

  // Search Module
  const filteredBooks = books.filter((book) =>
    book.title.toLowerCase().includes(search.toLowerCase())
  );

  // Reserve Module
  const reserveBook = (id) => {
    setBooks(
      books.map((book) =>
        book.id === id
          ? { ...book, reserved: true }
          : book
      )
    );
  };

  // Borrow Module
  const borrowBook = (id) => {
    setBooks(
      books.map((book) =>
        book.id === id && book.quantity > 0
          ? {

```

```

        ...book,
        borrowed: true,
        quantity: book.quantity - 1
      }
    : book
  )
);
};

```

// Return Module

```

const returnBook = (id) => {
  setBooks(
    books.map((book) =>
      book.id === id
        ? {
            ...book,
            borrowed: false,
            reserved: false,
            quantity: book.quantity + 1
          }
        : book
    )
  );
};

```

// Inventory Tracking Module

```

const addBook = () => {
  setBooks([
    ...books,
    {
      id: books.length + 1,
      title: newBook.title,
      author: newBook.author,
      quantity: Number(newBook.quantity),
      borrowed: false,
      reserved: false
    }
  ]);
};

```

```

setNewBook({
  title: "",
  author: "",
  quantity: ""
});
};

```

```

return (
  <div className="container">
    <h1>Library Management System</h1>

    <div className="card">
      <h2>Search Books</h2>

      <input
        type="text"
        placeholder="Search Books"
        value={search}
        onChange={(e) => setSearch(e.target.value)}
      />
    </div>

```

```

<div className="card">
  <h2>Admin Inventory Tracking</h2>

  <input
    type="text"
    placeholder="Book Title"
    value={newBook.title}
    onChange={(e) =>
      setNewBook({ ...newBook, title: e.target.value })
    }
  />

  <input
    type="text"
    placeholder="Author"
    value={newBook.author}
    onChange={(e) =>
      setNewBook({ ...newBook, author: e.target.value })
    }
  />

  <input
    type="number"
    placeholder="Quantity"
    value={newBook.quantity}
    onChange={(e) =>
      setNewBook({ ...newBook, quantity: e.target.value })
    }
  />

  <button onClick={addBook}>Add Book</button>
</div>

```

```

<h2>Books List</h2>

```

```

{filteredBooks.map((book) => (
  <div className="card" key={book.id}>
    <h3>{book.title}</h3>
    <p>Author: {book.author}</p>
    <p>Available Quantity: {book.quantity}</p>

    <button onClick={() => reserveBook(book.id)}>
      Reserve
    </button>

    <button onClick={() => borrowBook(book.id)}>
      Borrow
    </button>

    <button onClick={() => returnBook(book.id)}>
      Return
    </button>

    {book.reserved && <p>Book Reserved</p>}
    {book.borrowed && <p>Book Borrowed</p>}
  </div>
)}}
</div>
);
}

```

```
export default App;
```

src/index.js

```
import React from "react";
import ReactDOM from "react-dom/client";
import App from "./App";
import "./style.css";
```

```
const root = ReactDOM.createRoot(document.getElementById("root"));
root.render(<App />);
```

src/style.css

```
body {
  margin: 0;
  font-family: Arial;
  background: #f2f2f2;
}
```

```
.container {
  width: 80%;
  margin: auto;
  padding: 20px;
}
```

```
.card {
  background: white;
  padding: 20px;
  margin-bottom: 20px;
  border-radius: 10px;
}
```

```
input {
  width: 100%;
  padding: 10px;
  margin: 10px 0;
}
```

```
button {
  padding: 10px 15px;
  margin-right: 10px;
  background: blue;
  color: white;
  border: none;
}
```